

DATE: ___/___/___

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Data Structure Algorithm

Lab Tasks

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Infix $\longrightarrow$ Post-fix.

Convert Infix $\longrightarrow$ Postfix

Infix:

1. $A + B * (C^{\wedge} D - E)^{\wedge} (F + G * H) - I$

Postfix:-

- $CD^{\wedge}E-$
- FGH^*+
- $(CD^{\wedge}E-)(FGH^*+)^{\wedge}$
- $B(CD^{\wedge}E-FGH^*+)^{\wedge*}$
- $AB(CD^{\wedge}E-FGH^*+)^{\wedge*}+$
- $AB(CD^{\wedge}E-FGH^*+)^{\wedge*}+I-$

Postfix:-

$$ABCD^{\wedge}E-FGH^*+^{\wedge}+I-$$

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2. $A * (B + C * (D - E)) + F / G$

Prefix:

- DE -
- CDE - *
- BCDE - * +
- ABCDE - * + *
- FG /
- ABCDE - * + * FG / +

Evaluate Expressions :

1. $56 + 2 * 34 + -$

- $56 + \rightarrow 11$
- $112 * \rightarrow 22$
- $34 + \rightarrow 7$
- $227 - \rightarrow 15$

Result:-

15

2. Prefix: $- * 45 + 23$

- $* 45 \rightarrow 20$
- $+ 23 \rightarrow 5$
- $20 - 5 \rightarrow 15$

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Result:-

15

3 = Prefix : + * 234

• $+(23)4 \rightarrow (2*3)+4 \rightarrow 6+4=10$

Result:-

10

4 = Post fix : 93 * 42/-

• $93 * \rightarrow 27$

• $42 / \rightarrow 2$

• $27 - 2 \rightarrow 25$

Result:-

25